

LicenseGuard — Integration Guide

Vigno Smart Class · Game device-licensing for the download lane

How to make a game refuse to run on any machine except the one it was licensed for.

1. What LicenseGuard does

LicenseGuard is a single C# script you add to a Unity game. It runs automatically at startup and quits the game if it is not running on the licensed machine — so a copied, decrypted game folder will not run on another PC.

It closes the one remaining gap in the download-lane protection: while a game is running, its files are decrypted on disk and could be copied. With LicenseGuard compiled into the build, such a copy refuses to launch elsewhere.

Honest limit (state this to stakeholders)

Like all client-side DRM (Steam, Denuvo included), a determined reverse-engineer can disassemble the build and strip the check out. LicenseGuard stops essentially all casual copying and ties leaks to the buyer; it is not unbreakable. The only absolute guarantee is cloud-streaming (the files never reach the user).

2. How it works

- On Play, the Vigno Launcher decrypts the game and writes a signed token into <Game>_Data under a disguised filename (default: app.dat).
- The token is bound to the player PC's machine GUID and signed by the Vigno server with an RSA-2048 key.
- LicenseGuard runs BEFORE the first scene loads. It: (a) verifies the RSA signature, (b) checks the token's machine id == this PC's machine id, (c) checks the token is not expired.
- Any failure !' the game quits immediately. A copied folder carries a token bound to the original machine, so on any other PC the id will not match.

3. Prerequisites

- The Unity PROJECT (source) — not just the built .exe/zip. If you only have the built game, the original developer must do this.
- Unity 2021 LTS or newer, building a Windows Standalone (x86_64).
- The Vigno backend running, to fetch the public key.

4. Integration steps

Step 1 — Get your server's public key

Open this URL (server must be running) and copy the WHOLE block, including the BEGIN/END lines:

```
GET http://localhost:4000/.well-known/game-license-public-key
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQE... (several lines)
-----END PUBLIC KEY-----
```

Step 2 — Add the script

Drag game-sdk/LicenseGuard.cs into your project under Assets/Scripts/. Do NOT attach it to a GameObject — it runs automatically.

Step 3 — Configure the two values at the top of LicenseGuard.cs

```
const string LICENSE_FILE = "app.dat";    // must equal server LICENSE_GUARD_FILE

const string PUBLIC_KEY_PEM =
@"-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQE...
-----END PUBLIC KEY-----";
```

Step 4 — Set the .NET compatibility level (required)

Edit !' Project Settings !' Player !' Other Settings !' Api Compatibility Level !' .NET Standard 2.1. (Needed for RSA verification.)

Step 5 — Build the game

File !' Build Settings !' Windows (x86_64) !' Build. You get YourGame.exe + UnityPlayer.dll + YourGame_Data/ + MonoBleedingEdge/.

Step 6 — Re-upload to the platform

Zip the CONTENTS of the build folder (exe + _Data + dlls at the top level), then in the CMS open the same game item !' Upload (re-upload replaces the old file). Wait for: encrypting... !' file '.

Step 7 — Test it

In the launcher: Download !' Play on your PC !' it runs. Copy the running temp folder (%Temp%\vigno-game-...) to a second PC and run the .exe directly !' it quits ("not licensed for this device").

5. Troubleshooting (log messages)

LicenseGuard logs a clear reason if a launch is blocked. Check the build's Player.log or the launcher console.

- "No license found" !' the .exe was run directly, not through the Vigno Launcher.
- "This copy is not licensed for this device" !' machine GUID mismatch (the intended block on a copied game).
- "Invalid license signature" !' PUBLIC_KEY_PEM does not match the server key — re-copy it.
- "License expired" !' token older than GAME_LICENSE_TTL_DAYS; relaunch via the launcher to refresh.

Note: the guard is intentionally DISABLED inside the Unity Editor (no launcher/token there) so developers can keep testing. It only enforces in real builds.

6. Configuration reference

Server (backend/.env)

LICENSE_GUARD_FILE Disguised filename written into _Data (default app.dat). Must match the Unity script.

GAME_LICENSE_TTL_DAYS How long a token is valid (default 7). After this, the player must reopen via the launcher.

Unity script (LicenseGuard.cs)

LICENSE_FILE Must equal the server LICENSE_GUARD_FILE.

PUBLIC_KEY_PEM The public key from /.well-known/game-license-public-key.

Important: the signing key

The RSA keypair lives only in backend/keys/ (gitignored). Keep it backed up safely — if it is lost or regenerated, every game already built with the old public key will stop validating, and those builds must be rebuilt with the new key.

